Mortadha Ghnimi

AI Graduate Student & Research Assistant | Computer Vision, RAG, Remote Sensing

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EDUCATION

Grand Valley State University (GVSU) — M.S. in Artificial Intelligence (In Progress) 01/2026 – Present
Grand Rapids, MI

- Awarded a **merit-based Graduate Assistantship** (GA) at GVSU to pursue an M.S. in Artificial Intelligence.
- **Relevant Coursework**: Machine Learning, Data Mining, Computer Vision, Generative AI, MLOps, Cloud Application Development, high performance computing.

Faculty of Science of Bizerte — B.S. in Computer Science 09/2022 – 06/2025
Bizerte, Tunisia

- Graduated **top-ranked student** in the Faculty of Science of Bizerte across all years of the Bachelor's program.

EXPERIENCE

Graduate Research Assistant — Remote Sensing & Forest Disease Detection (Ongoing) 01/2026 – Present
Grand Valley State University MI, USA

- Conducting research on **forest and agricultural disease detection** using multispectral satellite imagery as a scalable alternative to drone-based monitoring.
- Analyzing spectral features across multiple bands to identify vegetation stress and health anomalies.
- Architecting data pipelines for **spatial** and **time-series** satellite imagery (NDVI) and establishing robust machine learning baselines for planned deep learning integration.
- Developed geospatial scripts in **Google Earth Engine** to visualize and pinpoint exact coordinates of vegetation stress anomalies.

AI & Automation Intern 05/2026 – 08/2026
Asjai Lou Casting Remote / New York, NY

- Turned a static casting-industry book into a connected digital platform by syncing Shopify/TikTok Shop purchases, fulfillment, customer accounts, and a continuously updated companion database.
- Automated secure post-purchase workflows using APIs and webhooks, including single-use account tokens, idempotent order processing, shipment tracking, admin content management, and error handling.
- Built a RAG assistant over the curated casting database for grounded natural-language search, along with the full-stack interface, backend APIs, testing, and deployment.

AI & Mobile Developer 01/2025 – 12/2025
Smart It Partner Tunis, Tunisia

- Integrated a **RAG-based chatbot and recommendation system** into a React Native application, delivering personalized workouts, meals, and user assistance through API-driven workflows.
- Developed a **deep learning computer vision system** for automated wildlife monitoring, classifying **10 forest animal species** with custom CNN architectures and data augmentation.
- Contributed to **API integration, debugging, prototyping, and application improvements**.

PROJECTS

1st Place Winner – Multiclass Image Classification Competition (Kaggle)
Python, TensorFlow, Keras, NumPy, Pandas

- Won 1st place by engineering a custom 3-CNN ensemble (WideResNet, ResNet-110) trained entirely from scratch.
- Achieved 93.4% accuracy on low-resolution datasets by utilizing advanced data augmentations (CutMix, Mixup, Cutout) and Cosine Annealing.
- Optimized final test predictions by executing a 3-round pseudo-labeling loop and 25-pass Test-Time Augmentation (TTA).

High-Performance Sparse Matrix Multiplication
C++, OpenMP, SIMD

- Ranked **#1 on the HPC leaderboard**, achieving **68.78× geometric-mean speedup** over baseline across six matrix widths with **6/6 correctness**.
- Optimized sparse matrix multiplication using packed CSR, OpenMP, SIMD, cache tiling, and prefetching on 296 MB of transcriptomics data.

Cost-Sensitive Healthcare Provider Fraud Detection
Python, Scikit-learn, Pandas, NumPy

- Built a healthcare fraud detection pipeline with **74 engineered provider-level features** from inpatient, outpatient, and beneficiary data.
- Evaluated LR, Random Forest, Gradient Boosting, and MLP models under class imbalance, reaching **0.946 ROC-AUC**.
- Applied cost-sensitive decision rules using reimbursement and investigation costs, cutting modeled screening cost by **~99% vs. baseline**.

GeoAgent — Agentic AI Research & Data Analysis Platform
Python, LangGraph, MCP, FastAPI, PostgreSQL, Docker

- Built an autonomous AI platform that plans and executes research workflows using LLMs, MCP tools, RAG, Python analytics, and SQL.
- Designed **planner–executor–verifier** workflows with persistent state, failure recovery, human approval, and grounded reporting.
- Evaluated the system across **50+ tasks**, measuring task completion, tool selection, grounding, latency, and inference cost.

SKILLS

- **Languages:** Python, C++, Java, JavaScript/TypeScript, SQL
- **AI & Machine Learning:** PyTorch, TensorFlow, scikit-learn, Hugging Face, Computer Vision, Recommendation Systems
- **GenAI & Agents:** LLMs, RAG, LangChain, LangGraph, MCP, Tool Calling
- **Data & Geospatial:** Pandas, NumPy, Google Earth Engine, PostgreSQL, pgvector
- **Backend & Systems:** FastAPI, REST APIs, Docker, Git, OpenMP, SIMD